

# E. KRISHNA DUTT

Hyderabad, Telangana, 500059 // Phone: +91-8143519298 // Email: krishnaemmadishetty@gmail.com

## PROFILE SUMMARY

Battery and EV systems professional with 12+ years of experience in automotive OEM and R&D environments, specializing in **high-voltage EV battery systems, hybrid ESS architectures, and vehicle-level integration**. Strong practitioner of **first-principles engineering, enabling root-cause driven problem solving and ground-up system design**. Proven technical leader with end-to-end ownership of battery system architecture, mechanical and electrical design, validation, and industrialization for buses, trucks, and hybrid applications. Extensive hands-on experience in **vehicle integration, including two-wheeler chassis, swing arm, and structural design**. Adept at leading cross-functional teams across design, testing, manufacturing, and suppliers to deliver robust, production-ready solutions. **Lean Six Sigma Green Belt certified**, driving continuous improvement, design robustness, and efficiency through Kaizen and structured problem-solving methodologies.

## KEY ACHIEVEMENT & TECHNICAL PROFILE

- **Battery Systems Design & Integration:** Mechanical architecture, packaging, and vehicle integration of cylindrical and prismatic LFP battery packs; end-to-end ownership from concept to production.
- Design & Developed the battery cooling base plate for 104.5 kWh battery pack in India.
- **Vehicle Integration & Structural Engineering:** Chassis, swing arm, and powertrain mounting design for two-wheelers, buses, trucks, and hybrid EV platforms.
- **Problem Solving & First-Principles Thinking:** Optimized a 30 kWh LFP battery pack, resolving weight, thermal, and packaging challenges using DFMEA, tolerance analysis, Pugh Matrix, DFMA, and VAVE methodologies to deliver an OEM-ready, validated design.
- **Design & Engineering Tools:** SolidWorks, Siemens NX (UG NX), PTC Creo, PLM systems, BOM management, APQP, DFMEA, DVP&R, PPAP, Engineering Pugh Matrix.
- **Manufacturing & Industrialization:** Tool & die design, tolerance stack-up, fixture and jig development, DFMA, and production-readiness engineering.
- **Leadership & Program Execution:** Cross-functional team leadership, strategic planning, project execution, vendor & supplier development, process standardization, people mentoring, and structured problem-solving for continuous improvement.

## LEADERSHIP & MANAGEMENT SKILLS

- Project & Team Leadership (cross-functional engineering teams)
- Strategic Planning & Execution (timelines, resource planning, cost control)
- Design Review & Decision-Making (technical feasibility, DFM, DFMEA)
- Vendor & Supplier Development (evaluation, quality assurance, cost negotiation)
- Process Standardization (proto checklists, review process, quality gates)
- People Development (skill matrix, mentoring, training, performance feedback)

## PROFESSIONAL EXPERIENCE

### **REPLUS Energy Pvt. Ltd., Pune**

#### **Deputy Manager – Product Design & Development | Aug 2025 – Present**

- Conducted design reviews and gate meetings ensuring design feasibility, manufacturability, and standards compliance.
- Led development of high-voltage battery packs (650V, 690Ah, 425kWh) for E-Bus (18M, 12M, 7M) and 55T truck applications.
- Led the 5MWh BESS container base – Rated Capacity -501596kWh, Nominal Voltage – 1331.2Vdc.
- Managed project planning, prototype builds, and validation testing.
- Coordinated with cross-functional teams to ensure smooth execution and test readiness.
- Handled resource planning, timelines, and cost tracking to ensure on-time delivery.
- Collaborated with validation teams to define acceptance criteria and test coverage.
- Ensured compliance with AIS-038 standards during validation.

### **CYGNI Energy Pvt. Ltd.**

#### **Manager – Product Development | Jan 2024 – Aug 2025**

- Led design of battery systems (2.6kWh to 30kWh) for 2W, 3W, 4W, and ESS applications.
- Designed enclosure for 5kWh ESS residential pack ensuring safety and manufacturability.
- Fabricated enclosure with minimal fixtures while achieving high dimensional and weld quality.
- Ensured AIS-156 standard compliance in all validation stages.
- Developed prototype checklists and validation processes, improving traceability.
- Reduced prototype cycle time by 20% through optimized assembly methods.
- Developed DFMEA, DVP, and APQP documents.
- Mentored junior engineers, reducing design rework by 25%.
- Led CFT teams and reported KPIs, risks, and design maturity.

### **Atumobile Pvt. Ltd., Hyderabad**

#### **Team Lead – R&D, Vehicle Building & Product Design | May 2021 – Jan 2024**

- Led complete product lifecycle as part of a startup team.
- Designed a Retro EV bike from concept to production including chassis and electrical integration.
- Conducted market study and cost analysis aiding strategic product selection.
- Guided chassis development including jigs and fixture creation.
- Managed ARAI certification process ensuring CMVR and AIS compliance.
- Handled vendor development for mechanical and fabricated components.
- Led proto builds, validation, and correlation tests.
- Coordinated cross-functionally to track progress and resolve technical issues.

### **Ather Energy Pvt. Ltd., Bangalore**

#### **Sr. Product Design Engineer – R&D | Jun 2017 – Apr 2021**

- Owned part design, validation, and release for Ather 450/450X vehicle components.
- Designed chassis, swing arm, gearbox casing, sheet-metal parts, and enclosures.
- Performed DFMEA, 2D drawings, and tolerance stack-ups.
- Led design reviews and validation plans with testing and reliability teams.
- Collaborated with electrical, battery, and integration teams for packaging and thermal management.
- Supported prototype builds with rapid design solutions.
- Worked with vendors on manufacturability and fixture design.
- Developed E-BOM and M-BOM structures.

**Escorts Ltd. (Agri Machinery Group), Faridabad****Design Engineer – R&D (Engine Design) | May 2016 – May 2017**

- Worked on engines for Power Track Euro 50 and Farm Track CRDI 75 tractors.
- Developed engine parts (cylinder block, oil sump, brackets) using Siemens UG NX.
- Created 3D models, 2D drawings, and layouts ensuring manufacturability.
- Performed fitment and assembly
- Benchmarked competitor engines for improvements.
- Coordinated assembly documentation with manufacturing teams.

Nucon Aerospace Pvt. Ltd., Hyderabad

**Jr. Engineer – Production Planning & Control | Sep 2014 – Apr 2016**

- Supported production schedules and workflow planning.
- Handled MRP planning and inventory control.
- Monitored stock levels aligned to production demands.
- Worked on ERP for dispatch tracking and BOM control.
- Maintained trackers and reports for production timelines.
- Supported process improvement initiatives.
- Gained experience in aerospace quality and documentation.

**Bharat Dynamics Ltd., Hyderabad****Intern – Production & Quality**

- Gained CNC operating and G-code programming experience.
- Assisted in quality inspection using precision tools.
- Supported tool setup and maintenance ensuring accuracy.
- Observed and practiced 5S methodologies.
- Learned manufacturing tolerances and inspection processes.

**ACADEMIC CREDENTIALS**

- Postgraduate (Tool Design) — Central Institute of Tool Design, Hyderabad (2015) — 61%
- B.Tech (Mechanical Engineering) — Bharat Institute of Engineering & Technology (2012) — 60%

**PERSONAL DETAILS**

- Date of Birth: 22/07/1990
- Current Residence: Hyderabad
- Languages: Telugu, English, Hindi

**DECLARATION**

I hereby declare that the information provided above is true to the best of my knowledge.